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Paper Code : OECCS701B Multimedia Systems

UPID : 007563

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) If an image has a width of 4 inch and an aspect ratio of 3 : 2, then what is its height?
- (II) What is generative graphics ?
- (III) What is raster graphics?
- (IV) MP3 produces three data rates from 96 Kbps to _____ Kbps.
- (V) List two type of multimedia.
- (VI) Which type of system is the PA system in audio devices?
- (VII) How many step processes are there to create a 3-D animation?
- (VIII) In Audio and Video Compression, voice is sampled at 8000 samples per second with _____
- (IX) What is the need for synchronization in multimedia presentations?
- (X) Define a Quad Tree.
- (XI) Give an example of a multimedia content design tool.
- (XII) For Music, we need to compress digitize signals at _____ i-mhz.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Write a note on Nyquist Theorem. [5]
3. Differentiate between formatted text and unformatted text. [5]
4. Write a short note on SGML. [5]
5. Define Sampling. What is meant by sampling rate? [5]
6. Why is quantization necessary? Distinguish between analog and digital waves. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) Compare different multimedia storage models and access methods with examples. [10]
- (b) Explain the role of buffering in maintaining multimedia synchronization. [5]
8. (a) How can you represent a digital image? [3]
- (b) Difference between captured image and stored image format. [5]
- (c) Explain different types of graphics format. [3]
- (d) Difference between sound and speech. [4]
9. (a) Explain in detail the components of a multimedia system with a neat block diagram. [10]
- (b) Discuss the application of multimedia. [5]
10. (a) Explain the basic concept of sound? How can you represent using computer? [7]
- (b) What are MIDI devices? [3]
- (c) Why MIDI devices are required in multimedia system? [5]
11. (a) Write a short note on CD-ROM. [5]
- (b) Define spatial and temporal redundancy. [5]
- (c) Write a short note on DVD. [5]

*** END OF PAPER ***